

AC Final Project

Learning Reduced-Round Cipher Behavior using Machine Learning

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Abstract. This report explores about whether machine learning models can actually predict as well as learn the relationships between the input which is the plaintext and the output which is the ciphertext for lightweight block ciphers with respect to the rounds. We tested four ciphers namely **SIMON, SPECK, PRESENT** and **GIFT** against four machine learning models namely Linear Regression, Multi-Layer Perceptrons (MLP), CNNs and Mutual Information Neural Estimation (MIME). The main aim is to increase the number of rounds until we reach an accuracy of about 50 percent. CNN outperformed the other models as it successfully learnt the highest number of rounds in three out of the four taken ciphers. PRESENT cipher turned out to be the easiest to crack as at round=3, the CNN hit an impressive bitwise accuracy of 98.8%. SPECK proved to be more robust as most models failed in the round 2 itself.

Keywords: Machine Learning · Round-Reduced Ciphers

1 Introduction

The block ciphers in the modern world are supposed to act like pseudorandom permutations, which means that the relation between the plaintext and the ciphertext should be almost impossible to figure out once one has performed sufficient number of rounds of confusion as well as diffusion. But, in case we are unable to use many rounds, then there is always a structure which a machine learning model might be able to pick up. The number of rounds depends on the robustness of the cipher. This project is about whether a model can predict the ciphertext for the corresponding plaintext for a specific round r after looking at a bunch of plaintexts and ciphertexts. We are testing it for four different cipher which belongs to either Feistel or SPNs against linear regression, CNNs, MLP and MIME.

2 Cryptographic Setup and Dataset

For the setup, we have gone with a black box oracle attack. We fixed one random secret key for each cipher and kept it constant across all the training and testing data sets. Also, we have taken plaintext uniformly from the $\{0, 1\}^n$ space and then generated the ciphertexts for $r=1, 2, 3, \dots$ rounds, and we continue till we get the desired threshold. For each pair of cipher and round, we generated about 100000 plaintext and ciphertext pairs and then used 80% and 20% split for training and validation set. It has been seeded at 42. The implementation of each cipher has been double checked using official test vectors to ensure the correctness of them.

For the machine learning use case, all the plaintext bits have been treated as the floating

vectors. Each output bit has been predicted using a sigmoid activation and corresponding loss is computed using Binary Cross Entropy (BCE). We used mainly three metrics namely Bitwise accuracy which is fraction of correct bits (random baseline of 0.5), block accuracy which is whether entire block is correct and mean Hamming distance which is average bit errors per sample (random base line= $n/2$).

Table 1: Detailed Specifications of Target Lightweight Ciphers

Cipher Model	Block (n)	Key (k)	Total Rounds	Design Architecture
SIMON-32/64	32-bit	64-bit	32	Feistel Structure (AND, ROT, XOR)
SPECK-32/64	32-bit	64-bit	22	ARX Design (ModAdd, ROT, XOR)
PRESENT-64/80	64-bit	80-bit	31	SPN (4-bit S-box + Bit Permutation)
GIFT-64/128	64-bit	128-bit	28	SPN (Bit-sliced Linear Layer)

3 Machine Learning Models

Four models were trained for every (cipher, round) combination. All neural models used Adam ($lr = 0.001$), cosine LR scheduling, batch size 4096, up to 40 epochs, and early stopping with patience 7. Experiments ran on Kaggle T4×2 GPUs.

3.1 Linear Regression (baseline)

We have trained one binary linear classifier per output bit using scikit-learn by also using L2 regularisation and LBFGS. We know that linear regression catches only affine relationships, so it can solve any round function made only with XORs or bit permutations, but if we involve an AND gate or say modular addition, then it will fall apart.

3.2 Multi-Layer Perceptron (MLP)

The setup which we made is of fully connected layers (512,512,256) with batch normalisation, ReLU as a non-linear activation and a 10% dropout. This is a good model as it has a global view of all the input bits at once. It is advantageous if the cipher's structure spans the whole block, but due to this and relative to CNN, it requires more parameters.

3.3 Convolutional Neural Network (CNN)

We have used three 1 dimensional convolutional blocks with channels (64,128,256), with kernel size 3 and then followed by two dense layers (512 and 256). The CNN is often a perfect fit for this project because of its inductive bias which matches with how cipher works. For instance, S Box operates on small 4 bit nibbles and ARX functions link the adjacent bits via carry propagation.

3.4 Mutual Information Neural Estimator (MINE)

We have used MINE to maximise the lower bound of the mutual information $I(X;Y)$ between the plaintext and ciphertext. It has been adapted into a generative predictor with a shared [512,512,256] encoder and an autoregressive decoder. If the mutual information is no longer there, then no model is going to get better than random guessing.

4 Experimental Results

4.1 SIMON-32/64

SIMON uses a Feistel structure where we know that the only non linear part is a bitwise AND. At $r=1$, where r is round number, all the neural models reached almost 100% accuracy. The Multi-Layered Perceptron proved to be resilient as it had 95.2% at $r=2$ and 61.3% for $r=3$. CNN and MINE gave up after the second round itself.

The good part is that linear regression hits an accuracy of 75% at $r=1$ because XOR and rotation operations are affine, but at $r=2$, it dropped to 50%. Thus, it is sufficient to explain that an AND gate per round is sufficient to stop any cracks using its linear nature. For the Hamming Distance metric for CNN, $r=1$, the errors are almost zero, but at $r=4$, the distribution looks like Binomial(32,0.5).

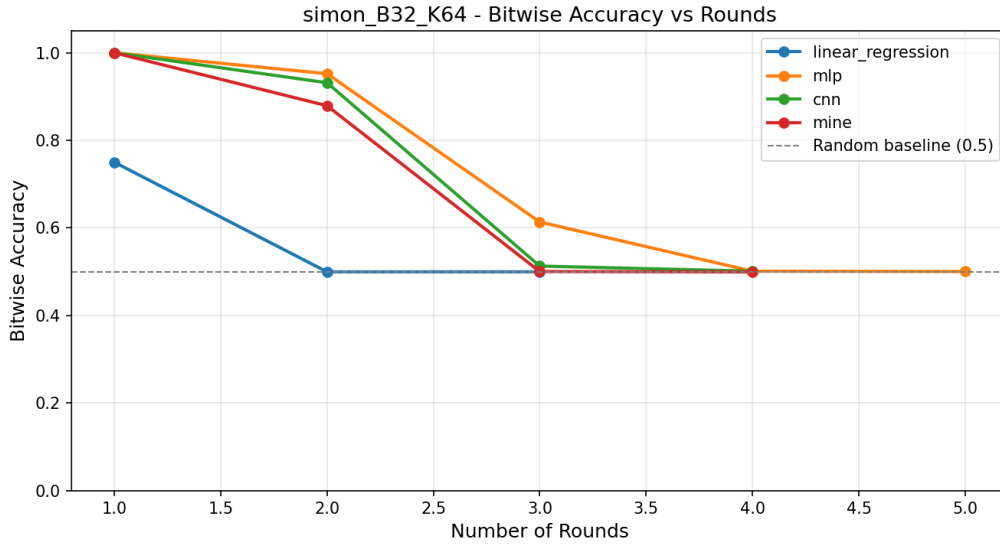


Figure 1: SIMON-32/64: bitwise accuracy vs. round count for all four models.

Table 2: Bitwise Accuracy Performance on SIMON-32/64

Model Class	$r = 1$	$r = 2$	$r = 3$	$r = 4$	Learned r
Linear Regression	75.0%	49.9%	50.0%	NA	1
MLP	100%	95.2%	61.3%	50.1%	3
CNN	100%	93.1%	51.3%	50.1%	2
MINE	100%	87.9%	50.0%	49.9%	2

Figure 2 shows the Hamming distance distribution at $r = 1$ and $r = 4$ for the CNN model. At $r = 1$ hamming distance is near zero; by $r = 4$ the distribution matches a Binomial(32, 0.5) which is random guessing.

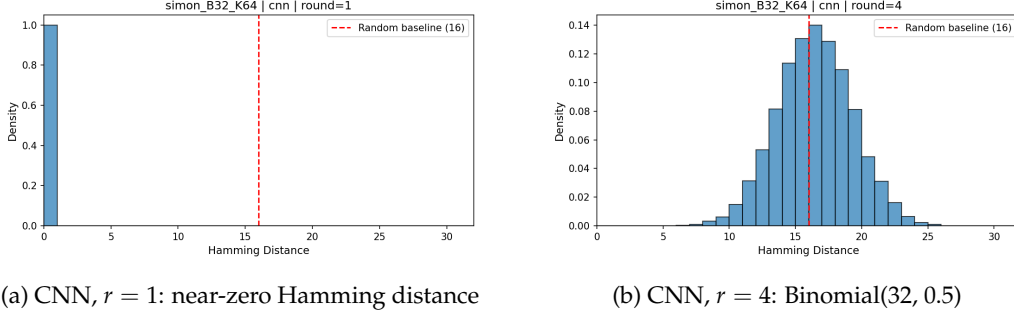


Figure 2: SIMON-32/64 CNN Hamming distance distribution at $r = 1$ (left) and $r = 4$ (right).

4.2 SPECK-32/64

SPECK is a cipher which swaps out the AND gate for modular addition. Carry bits in the modular addition create much more complex interaction of bits compared to the simple logic gates. As per our results, at $r=1$, linear regression is at the verge of randomness (49.9%), which means that a single round of SPECK is enough to completely destroy any linear model.

The models like CNN, MLP and MINE were able to learn and achieve fairly well for the round one. CNN hits at 97.5%. But the robustness and resistance of this cipher is very high that at $r=2$, all the models crashed back to random baseline of 50%. Out of all the ciphers taken by us, SPECK proved to be very resistant to ML models.

Table 3: Bitwise Accuracy Benchmarks for SPECK-32/64

Model Type	Round 1	Round 2	Round 3	Effective Limit
Linear Regression	49.9%	50.0%	NA	0 Rounds
MLP	97.9%	50.0%	50.1%	1 Round
CNN	97.5%	49.9%	50.0%	1 Round
MINE	84.5%	49.9%	50.1%	1 Round

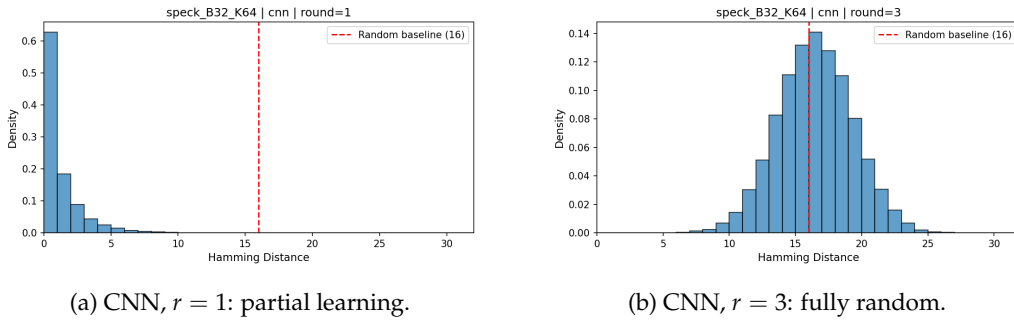


Figure 3: SPECK-32/64 CNN Hamming distance distribution at $r = 1$ and $r = 3$

4.3 PRESENT-64/80

PRESENT is the most learnable cipher we studied, largely due to it using a simple bit-shuffle to achieve diffusion. However, it also takes significantly longer to encrypt a

message thoroughly due to the bits not being mixed properly within fewer rounds as a result.

PRESENT has a fixed key and an $r=1$ transformation that find the ciphertext through an affine map via an S-box lookup with a bit permutation. Consequently, all four models achieve 100% at $r=1$. CNN sustains this in $r=2$, achieves 98.8% at $r=3$, and collapses below the 50% mark at $r=4$. MLP and MINE both collapse after $r=2$, and linear regression fails faster because it loses track of the linear structure.

Table 4: Bitwise Accuracy Benchmarks for PRESENT-64/80

Model Type	Round 1	Round 2	Round 3	Round 4	Round 5	Effective Limit
Linear Regression	100%	65.5%	54.0%	51.1%	50.4%	1 Round
MLP	100%	99.99%	62.7%	51.2%	50.3%	2 Rounds
CNN	100%	100%	98.8%	52.3%	50.3%	3 Rounds
MINE	100%	99.99%	64.0%	51.2%	50.3%	2 Rounds

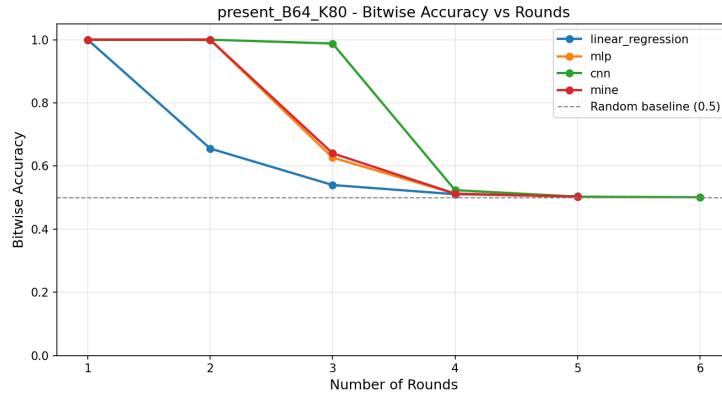


Figure 4: PRESENT-64/80: bitwise accuracy vs. round count for all four models

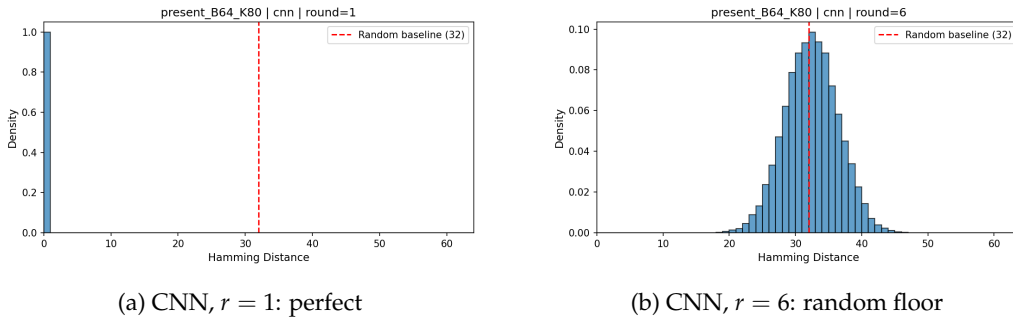


Figure 5: PRESENT-64/80 CNN Hamming distance distribution at $r = 1$ and $r = 6$

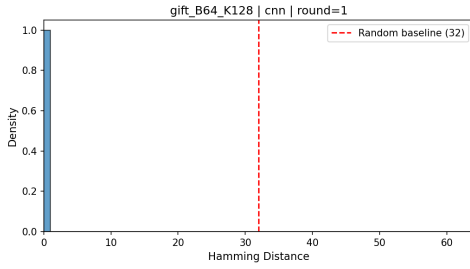
4.4 GIFT-64/128

GIFT is a modification of the PRESENT cipher that also works with a 4-bit S-box but opts for a bit-sliced linear layer over simple shuffling to achieve diffusion faster. Linear regression performs poorly as it achieves only 59.3% at $r=1$. This means the affine structure is immediately destroyed with this modification. MLP and MINE only succeed in the

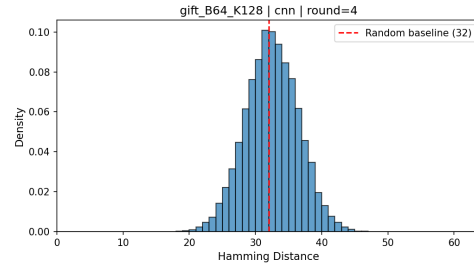
first round before dropping to near-random values, and while CNN performs well in the first two rounds with a near perfect accuracy (99.9% at $r=2$), it also collapses by the third round. Hence, GIFT is significantly stronger than PRESENT with its modification.

Table 5: Bitwise Accuracy Benchmarks for GIFT-64/128

Model Type	Round 1	Round 2	Round 3	Round 4	Effective Limit
Linear Regression	59.3%	50.0%	50.0%	NA	1 Round (partial)
MLP	100%	57.7%	50.0%	50.0%	1 Round
CNN	100%	99.9%	51.5%	50.0%	2 Rounds
MINE	100%	56.3%	49.9%	49.9%	1 Round



(a) CNN, $r = 1$: almost perfect



(b) CNN, $r = 4$: random floor

Figure 6: GIFT-64/128 CNN Hamming distance distribution at $r = 1$ and $r = 4$

4.5 Cross-Cipher Comparison

Table 6: Maximum learnable rounds per cipher and model

Cipher	Linear Regression	MLP	CNN	MINE	Best Result
SIMON-32/64	1	3	2	2	3 rounds (MLP)
SPECK-32/64	0	1	1	1	1 round (MLP, CNN, MINE)
PRESENT-64/80	1	2	3	2	3 rounds (CNN)
GIFT-64/128	1 (59.3%)	1	2	1	2 rounds (CNN)

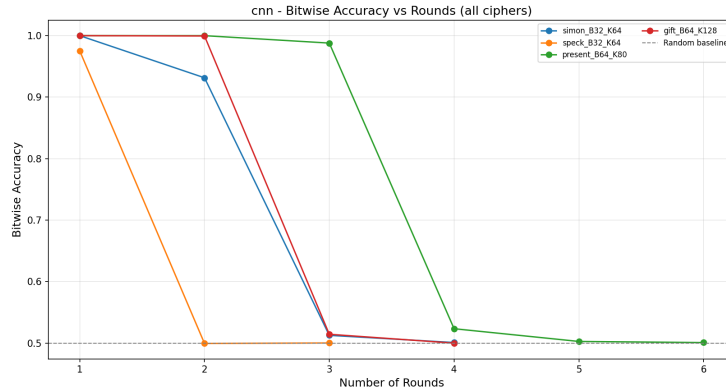


Figure 7: CNN model across all four ciphers; PRESENT>SIMON/GIFT>SPECK

5 Limitations

A single key was used for each cipher, allowing the models to learn key specific mappings to decrypt them. Further, only four ciphers were used to evaluate these models, all of which were block ciphers. Additionally, using larger or more transformer-based models might help extend the learnability by one or two extra rounds, and more training data could make the model more accurate nearer its limit.

6 Conclusion

Our project shows how ML models can be trained to predict the output of block ciphers with high accuracy for a low number of rounds. The key takeaways of the project are as follows:

1. CNN is significantly better than the other models in decrypting the patterns used to construct nearly all the ciphers we analysed. We have determined that this is due to the inductive bias carried by its 1D convolution layers. Also, if diffusion in two rounds varies only by a few bits, then MaxPooling layers are able to detect it due to spatial invariance.
2. The only exception to the objective advantage CNN has in our analysis is the case of the SIMON cipher, in which CNN is beaten out by MLP by one round, possibly due to its low-diffusion Feistel structure that complements MLP's design that is built to focus on all bit positions.
3. The order of increasing learnability (PRESENT>SIMON/GIFT>SPECK) is primarily determined by the diffusion speed of the cipher, i.e. the number of rounds required for a change in one input bit to affect all output bits with a probability nearing 0.5. PRESENT is the easiest cipher because it only applies a permutation to the bits which needs several rounds before diffusion is enough. GIFT uses a better permutation than PRESENT which reaches sufficient diffusion quicker. So it is harder to learn than PRESENT. SIMON uses AND and rotation in its Feistel structure. It is roughly as easy as GIFT. SPECK is the hardest cipher because modular addition gets rid of affine structure instantly.
4. Linear regression makes the order of learnability clear. For it, PRESENT is fully learnable in the first round due to the fixed key round being an affine map. SIMON only has 75% accuracy at round 1 because it is less affine than PRESENT. Finally, SPECK and GIFT have accuracy equal to random guessing even at round 1 because they are non-linear.
5. MINE performs almost as well as MLP, but it is marginally less accurate due to the decoding process being less efficient. Importantly, it loses track of information at the same round that accuracy drops significantly across all ciphers, confirming the poor performance is the result of successful pseudorandom diffusing algorithms and not a weak model. Consequently, full-round cipher outputs are effectively random from an ML perspective.

Thus, the tested ciphers are confirmed to be adequately secure to ML-based attacks.

7 References

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